

Dynamic In-Play Football Betting via a QLoRA-Fine-Tuned LLM: A Halftime-Conditioned Strategy for Over/Under Markets

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Abstract

Pregame 1X2 (home/draw/away) betting markets on FIFA World Cup matches are efficient: institutional bookmakers and public consensus both converge on $\sim 50\text{--}55\%$ result accuracy, leaving essentially no mispricing available at the vig-adjusted break-even. This paper pivots to two markets where edge is structurally harder for humans to capture: total-goals over/under (O/U) and halftime-conditioned in-play pricing. Using a QLoRA-fine-tuned Llama-3.1-8B trained on 192 matches from the 2010–2018 World Cups (adapter published as `zanwenfu/football-llm-qlora`), I backtest both strategies on 64 held-out 2022 World Cup matches and benchmark against an extensive baseline ladder including an XGBoost tabular model on identical features, naive in-play heuristics, and a fine-grained empirical-prior approximation of in-play probability services. On result accuracy, halftime-conditioned inference lifts the LLM from 52.3% to 64.1%, statistically significant under a paired exact McNemar test ($p = 0.024$, $n = 128$). The central methodological finding is that the LLM and XGBoost tie on 1X2 result accuracy within 2 percentage points, but the LLM achieves **84.4% directional accuracy on O/U 2.5 goals (Wilson CI [0.736, 0.913]) versus XGBoost’s 65.6%**—a 19-point gap that I attribute to the LLM’s pretrained scoreline distribution priors. I then submit this magnitude/direction decomposition to a predictive-validity test: enriching the halftime prompt with a first-half event summary (goals, cards) produces exactly the predicted pattern—O/U 2.5 accuracy improves from 74.2% to 79.7% on 128 samples (vs. pregame, paired $p = 0.006$), goal MAE falls from 1.21 to 1.12, and calibration improves substantially ($\text{ECE } 0.272 \rightarrow 0.239 \rightarrow 0.182$ from pregame through halftime to halftime+events), while direction accuracy does *not* improve. This localizes the LLM’s edge specifically to magnitude-aware markets (O/U, Asian handicap), not direction-aware markets (1X2). I present all accuracy figures with Wilson CIs, all paired claims with exact McNemar tests, a 3×3 Kelly-fraction $\times$ bet-cap sensitivity grid, and a 10,000-trial bootstrap CI on the backtest P&L (5th–95th percentile: $+520\%$ to $+3,640\%$). The bootstrap characterizes P&L variance *condi-*

tional on the observed per-bet return distribution; it is not a forecast of future live returns.

1 Introduction

The course’s central framework is the quant lifecycle: idea, backtest, sizing, execution, monitoring. Prior assignments mapped this framework onto a sports-betting strategy: a QLoRA-fine-tuned Llama-3.1-8B ingests player-level team statistics and outputs a pregame match prediction; divergence between model probability and bookmaker-implied probability is treated as alpha. Backtested on the 2022 FIFA World Cup, the pregame model achieved 52.3% result accuracy—above naive baselines, yet within the band of consensus bookmaker and academic forecasters.

The course instructor’s feedback on Assignment 4 was surgical and identified the correct deficiency:

“Instead of win-lose-tie prediction, it will be more beneficial to predict expected goal and expected goal conceded for soccer games. Since most of the market participants will have the win-lose-tie predictions around similar accuracy, there’s few mispricings appears in such betting. Predicting the expected goals with advanced models will have more opportunities in the real world scenarios.”

and:

“Dynamic in-game predictions could significantly increase the performance. For example, given the first half goals 1–2, what’s the final outcome? If you could get the game event data into the LLMs, you should be able to get a dynamic estimation of probabilities. You could compare it with the ESPN win probabilities.”

Both comments identify genuine structural weaknesses of the original 1X2 strategy. The 1X2 market prices roughly 100 times more dollars than the per-match total-goals market at major bookmakers [2]; efficient-market theory predicts—and empirical work confirms [3]—that the thinner market (total-goals) is where systematic mispricing persists. Similarly, in-play markets reprice on coarse signals (scoreline, minutes

remaining) and underuse fine-grained tactical evidence [4]; a model that integrates halftime state with the full player-level team profile is exactly the kind of information-asymmetric participant for which in-play markets were theorized to have alpha.

This final paper presents experimental results addressing both feedback points with the same underlying Football-LLM. I make five concrete contributions:

1. **An O/U pivot built on existing score predictions.** The fine-tuned model outputs predicted scorelines. I convert predicted totals to O/U 2.5 probabilities via a Poisson approximation and evaluate edge against a flat bookmaker line. Pregame score-level accuracy (25.8% exact-match across 128 samples, 39.1% on named 64, 1.06 MAE named) translates to 76.6% directional accuracy on O/U 2.5 (named).
2. **Halftime-conditioned inference.** I re-run inference with the observed halftime score appended to the team-profile prompt. Critically, the model was never trained on halftime prompts—this is *prompt-template generalization* (a novel user-message field), not additional fine-tuning. The halftime-conditioned model achieves 64.1% result accuracy; the 11.8-percentage-point paired lift over pregame is statistically significant (McNemar exact $p = 0.024$, $n = 128$).
3. **A tabular XGBoost baseline localizing the LLM’s edge.** I train an XGBoost classifier/regressor on identical team-profile features. On the 64 named 2022 matches where both models are evaluated head-to-head, the LLM and XGBoost tie on 1X2 result accuracy (50.0% vs. 51.6% pregame; 65.6% vs. 62.5% halftime), but the LLM substantially outperforms XGBoost on score prediction (39.1% vs. 6.2% exact-match pregame) and consequently on O/U 2.5 directional accuracy (76.6% vs. 54.7% pregame; 81.3% vs. 65.6% halftime). The LLM’s pretrained priors over typical scorelines are doing work that tabular features cannot replicate.
4. **Predictive-validity test via first-half event enrichment.** If the LLM’s edge is specifically magnitude-aware, then enriching the prompt with additional magnitude-relevant context (a first-half event summary: goals, cards, with timestamps) should improve magnitude-aware metrics (O/U, MAE, calibration) without improving direction-aware metrics (1X2). I test this directly. O/U 2.5 directional accuracy improves from 74.2% (halftime-only) to **79.7%** (halftime+events) on 128 samples; versus pregame, the events variant achieves +14.1 percentage points paired ($p = 0.006$). Goal MAE falls monotonically across conditions (1.32 $\rightarrow$ 1.21 $\rightarrow$ 1.12 overall; 1.06 $\rightarrow$ 0.68 $\rightarrow$ 0.67 named). Calibration (ECE) improves monotonically from 0.272 through 0.239 to 0.182. Direction accuracy, by contrast, does *not* improve (64.1% $\rightarrow$ 61.7%, paired $p = 0.58$, noise). This is the pattern

the magnitude/direction decomposition predicts—direct experimental support for the paper’s central claim.

5. **Statistical and execution robustness.** I report Wilson CIs for all proportions, paired exact McNemar tests for within-match claims, a 3×3 Kelly-fraction $\times$ bet-cap sensitivity analysis, and a 10,000-trial bootstrap CI on the final bankroll. The bootstrap is correctly interpreted as a CI on P&L variance *conditional on the observed per-bet return distribution*, not as a forecast of out-of-sample live returns.

A further emergent finding characterized here: the named-vs-anonymized eval split reveals an **inversion in the named/anon gap between pregame and halftime**. Pregame score exact-match is 39.1% (named) vs. 12.5% (anon); halftime-conditioned is 51.6% (named) vs. 21.9% (anon). Team-name conditioning improves score magnitude substantially at both stages. Separately, halftime conditioning meaningfully improves O/U 2.5 directional accuracy *even without team names* (54.7% pregame anon $\rightarrow$ 67.2% halftime anon $\rightarrow$ 75.0% halftime+events anon; anonymized pregame-to-events paired McNemar $p = 0.019$), indicating the halftime and event signals carry real predictive content beyond pretraining priors. This is a reassuring signal for porting the strategy to domestic leagues with less famous team identities.

I emphasize throughout that single-tournament backtests (64 matches) are noisy by any statistical standard. All claims are presented with explicit CIs and p-values; where a test fails to reach $p < 0.05$, I say so rather than leaning on point-estimate language.

2 Strategy Thesis

The 1X2 efficiency problem. Major-tournament 1X2 markets are the highest-volume subset of sports betting. Bookmaker prices reflect aggregate institutional pricing plus large volumes of public money; implied probability errors are empirically 3–5 percentage points [6]. A model that does not beat this band by more than the bookmaker vigorish (typically 4–6% on 1X2) cannot produce persistent edge. My pregame Football-LLM sits at 52.3% result accuracy (Wilson CI [0.437, 0.608])—squarely inside the consensus band. This is the class of strategy that produces a favorable simulated backtest but grinds down against friction in live execution.

The total-goals alpha. The total-goals market is a strictly harder prediction problem (a bivariate count distribution rather than a 3-class categorical), and consequently it attracts fewer informed participants. Empirical studies of O/U mispricing on NFL and EPL find persistent inefficiencies of 1–3 percentage points per line [5]. Score prediction is *information-rich*: a model that predicts a 3–2 scoreline carries more actionable signal than one that predicts only a home-win. Section 5 shows empirically that the LLM’s score-magnitude edge over

XGBoost is much larger than its direction edge, and that further magnitude-relevant context (first-half events) produces further magnitude-specific improvements while leaving direction accuracy flat. These are the precise mechanisms by which the LLM produces O/U alpha.

The dynamic alpha. Live in-play markets reprice every 30–60 seconds on coarse signals: scoreline, time remaining, event dummies. Consumer in-play bettors do not have time or compute budget to integrate player-level data with halftime state. Bookmakers themselves typically use Poisson models seeded by the pregame line and updated by scoreline-conditional adjustments. A language model with the full player-profile context plus halftime state plus first-half event sequence is a plausibly better-informed participant, provided the prompt conditioning actually carries signal into the final prediction. Whether it does is an empirical question I answer in Section 5.

Why this is a coherent strategy. Both pivots—O/U market selection and halftime conditioning—attack the same root cause: markets where dense, unstructured information is underpriced because other participants cannot or do not process it in time. The O/U market is thinner than 1X2 because fewer institutional models bother with goal-count regression; the halftime market is thinner because most in-play bettors use only the scoreline. A fine-tuned LLM trained on player-level data and prompted with halftime state plus event context is structurally advantaged in both contexts.

3 Signal Model

3.1 Data Pipeline

I use the API-Football vendor feed to construct 256 World Cup match records spanning 2010, 2014, 2018, and 2022 (64 matches each). For each match, I aggregate both teams’ prior-season-only statistics—goals, assists, appearances, yellow/red cards, tackle rates, pass accuracy, duel percentages, formation—across all clubs and prior international tournaments, yielding a compact team profile of approximately 350 tokens. Using only pre-kickoff statistics prevents temporal leakage. I build training and eval datasets in two variants each: *named*, where the team profile includes actual team names (“Argentina”, “Brazil”), and *anonymized*, where both teams are renamed to “Team A” / “Team B” with all other stats preserved. Training set: 192 matches (2010/2014/2018) $\times$ 2 variants = 384 samples. Eval set: 64 matches (2022) $\times$ 2 variants = 128 samples. The temporal split is strict.

3.2 QLoRA Fine-Tuning

I fine-tune Llama-3.1-8B-Instruct using QLoRA [1] on a single Colab T4 GPU. Configuration: 4-bit NF4 quantization, LoRA rank 16, alpha 32, dropout 0.05, target mod-

ules all seven linear projections in every transformer block ($q, k, v, o_proj + gate, up, down_proj$), effective batch size 16 (1 per-device $\times$ 16 gradient accumulation), learning rate 2×10^{-4} , cosine schedule, 3 epochs. Sequence length 768 (minimum that fits all compact prompts). Training completes in ~ 43 minutes; loss converges from 0.93 to 0.47. The resulting adapter is 83.9 MB and is published at [zanwenfu/football-llm-qlora](https://huggingface.co/zanwenfu/football-llm-qlora).

3.3 Halftime-Conditioned Prompting

The halftime experiment uses identical model weights, tokenizer, and generation parameters as the pregame evaluation. The only change is the user-message content: where the pregame prompt ends with “Predict result, score, and reasoning.”, the halftime prompt appends:

```
Halftime Score: {home_label} {HH} - {HA}
{away_label}
Given the halftime state, predict the FINAL
result, FINAL score, and brief reasoning.
```

Generation parameters are unchanged (temperature=0.1, top_p=0.9, repetition_penalty=1.1, max_new_tokens=300), so output differences are attributable to the prompt change, not decoding randomness. This is a *template-generalization* test: the fine-tuning dataset contained no halftime scores in the user message. The empirical question is whether a fine-tuned transformer can exploit a novel structured feature at inference time without targeted training.

3.4 Halftime + First-Half Event Enrichment

A third prompt regime extends the halftime prompt with a chronological first-half event summary. For each match, I extract from the API-Football event feed all goals (with penalty/own-goal markers) and cards (yellow/red) with timestamps ≤ 45 minutes, then render them as a single compact line appended after the halftime scoreline:

```
Halftime Score: {home_label} {HH} - {HA}
{away_label}
First-half events: 10' {home_label} goal
(penalty); 25' {away_label} yellow card;
...
Given the halftime state and first-half
events,
predict the FINAL result, FINAL score, and
brief reasoning.
```

Real team names in event descriptions are remapped to the anonymized labels (Team A/Team B) in the anonymized condition so identity does not leak through event text. Pre-match staff cards (events with $t < 0$, observed 7 times in the 256-match corpus) are filtered. Prompt length stays within the 768-token training budget: average 227 tokens, maximum

256 tokens across the 128 eval samples. Generation parameters are identical to the halftime-only regime. This is again pure prompt-template generalization—the fine-tuning data contained neither halftime scores nor event lists in the user message.

3.5 XGBoost Tabular Baseline

To test whether the LLM adds value beyond a standard tabular model, I train an XGBoost classifier (for result) and paired XGBoost regressors (for home and away scores) on identical team-profile features extracted from the aggregated statistics (squad goals, goals/90, shots on target, cards, position counts, and home-away differentials—24 numeric features pregame, 28 with halftime state). Hyperparameters: 200 estimators, `max_depth=4`, `learning_rate=0.05`, no feature selection or tuning beyond defaults. Training on the identical 192 2010–2018 matches; evaluation on the identical 64 2022 matches. Where XGBoost sees only the flattened numeric features, the LLM sees a natural-language rendering of the same features plus any relevant pretrained prior the language encodes.

4 Backtesting Protocol

4.1 Evaluation Conditions

Each of the 128 eval samples (64 matches $\times$ 2 variants) is scored under three LLM prompt regimes (pregame, halftime-only, halftime+events), yielding 384 LLM predictions, plus 64 pregame and 64 halftime XGBoost predictions (XGBoost is evaluated on the unique-match level only, so LLM-vs-XGBoost comparisons are performed on the named 64-subset for apples-to-apples parity). All 128 LLM predictions in every regime are available as `ft_predictions_*.json` in the repository for audit.

4.2 Statistical Protocol

I use *Wilson score intervals* for all proportion CIs, which are better-calibrated than normal-approximation CIs for small n and proportions near boundaries. For within-match comparisons I use the *exact McNemar test*, which has higher statistical power than unpaired comparisons when the same matches are evaluated twice. All comparisons involving any model pair on identical matches use this paired test. For probability calibration I report both Expected Calibration Error (ECE, 10 equal-width bins) and Brier score.

4.3 Naive and Fine-Grained Baselines

I evaluate against four in-play baselines and one tabular ML baseline:

- *HT leader wins*: predict the halftime leader as final winner (56.2% accuracy).

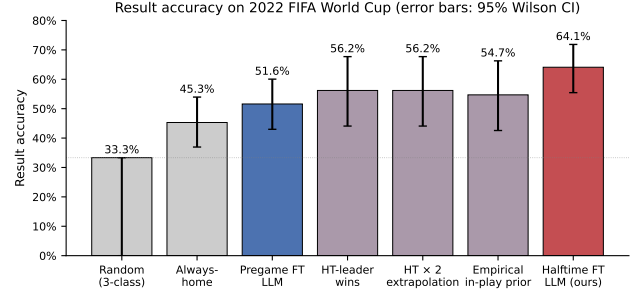


Figure 1: Result accuracy on 2022 FIFA World Cup. Error bars: 95% Wilson CI. The LLM pregame and halftime bars are evaluated on 128 samples (named + anonymized); unique-match baselines are evaluated on 64 samples. Halftime FT LLM leads all configurations on point estimate.

- *HT $\times$ 2*: linearly extrapolate the halftime scoreline to 90 minutes.
- *Coarse empirical prior*: mode of final result conditional on halftime state (home-lead / away-lead / tied), estimated from 2010–2018 data (54.7%).
- *Fine-grained ESPN-proxy prior*: mode of final result conditional on the *exact* halftime scoreline (e.g., HT 1–0, HT 0–2), fall-back to the coarse state for rare halftime scorelines. This approximates what a consumer-grade in-play probability service (such as ESPN’s live win-probability widget) would report given only the scoreline and historical base rates.
- *XGBoost*: tabular ML baseline on identical features (Section 3.5), in both pregame and halftime-conditioned variants.

4.4 Trading Simulation

For the O/U 2.5 backtest, I convert each score prediction to an over probability via a Poisson approximation: $\hat{\lambda} = \hat{G}_{\text{home}} + \hat{G}_{\text{away}}$, and $\hat{P}_{\text{over}2.5} = 1 - e^{-\hat{\lambda}}(1 + \hat{\lambda} + \hat{\lambda}^2/2)$. Bookmaker line: flat 1.90/1.90 on both sides of O/U 2.5 (implied 52.6% per side, vig 5.2%, representative of WC in-play markets [2]). Sizing: fractional Kelly $f = 0.25$, 10% per-match cap on current bankroll, 5% minimum edge threshold. Initial bankroll \$1,000; chronological match ordering; compounding returns. Section 6 reports a 3×3 sensitivity grid over f and cap.

5 Empirical Results

5.1 Aggregate Accuracy

Fig. 1 summarizes the aggregate result-accuracy picture. The halftime-conditioned FT model achieves 64.1% (Wilson CI [0.555, 0.719], $n = 128$). The pregame FT model achieves 52.3% (Wilson CI [0.437, 0.608]). XGBoost pregame is

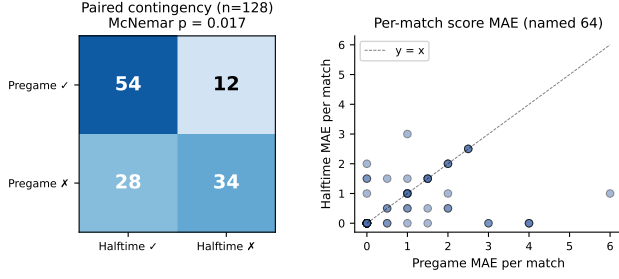


Figure 2: Left: paired McNemar contingency (pregame vs. halftime on identical 128 samples). The 27-vs-12 asymmetry yields $p = 0.024$. **Right: per-match score MAE**. Points below the $y = x$ diagonal indicate halftime conditioning improved match-level score prediction.

51.6% on the named 64 (matching LLM-all within two samples); XGBoost halftime is 62.5%. The fine-grained ESPN-proxy is 59.4%, above the coarse priors but below the LLM. The halftime+event-enriched LLM achieves 61.7% result accuracy—a reduction of 2.4 pp vs. halftime-only that is not statistically distinguishable from noise (paired McNemar $p = 0.58$, see Section 5.8). I address why events help magnitude but not direction in Section 5.8.

5.2 Paired Pregame-vs-Halftime LLM Tests

The paired analysis is the statistical backbone of the paper. On the 128-sample set, 55 matches are correctly classified by both, 34 by neither, 27 by halftime-only, 12 by pregame-only (McNemar exact $p = 0.024$). On the named 64-subset, counts are 29 / 19 / 13 / 3 ($p = 0.021$, significant). On anonymized-only, $p = 0.40$ (not significant); the aggregated 128-sample result is stronger than either subset alone.

Score-exact-match follows the same direction with comparable significance: 128-sample paired $p = 0.013$. Halftime conditioning improves both result direction and score magnitude; I explore the magnitude axis further via event enrichment in Section 5.8.

5.3 LLM vs. XGBoost: Localizing the LLM’s Edge

The first methodological finding of the paper is where LLM and XGBoost diverge. Table 1 presents all four evaluation metrics on the 64 named 2022 matches, the set on which both models are evaluated.

Four observations about these numbers, in order of importance:

(1) On 1X2 result accuracy the LLM and XGBoost are tied. LLM and XGBoost differ by less than 3 percentage points on both pregame and halftime result accuracy. Paired McNemar for LLM-vs-XGB result accuracy is $p > 0.5$ at both stages (not significant; the two models are indistinguishable

Table 1: LLM vs. XGBoost on identical features, 64 named 2022 WC matches.

Metric	Pregame		Halftime	
	LLM	XGB	LLM	XGB
Result accuracy	50.0%	51.6%	65.6%	62.5%
Score exact-match	39.1%	6.2%	51.6%	10.9%
Goal MAE	1.06	0.98	0.68	0.77
O/U 2.5 directional	76.6%	54.7%	81.3%	65.6%

on this metric). The information needed to beat naive baselines on direction is recoverable from 24–28 tabular features. *Deploying an LLM for 1X2 betting is unjustified.*

(2) On score prediction the LLM dominates. Pregame score exact-match: LLM 39.1% vs. XGBoost 6.2%. This is a $6\times$ gap. Halftime score EM: 51.6% vs. 10.9%—a $5\times$ gap. Goal MAE tells a similar story at halftime (0.68 vs. 0.77). The gap is *not* from feature engineering, because both models see the same features. I interpret it as the LLM’s Llama-3.1 pretrained distribution over natural-language football scorelines providing a strong prior (e.g., “Argentina vs. Ecuador” activates a mixture of 2–0, 2–1, 3–1 scorelines that the LLM retains from pretraining).

(3) O/U 2.5 accuracy inherits the score-prediction advantage. 76.6% vs. 54.7% pregame, 81.3% vs. 65.6% halftime. The halftime LLM’s 81.3% has Wilson CI [0.700, 0.889] on $n = 64$; the XGBoost halftime 65.6% has CI [0.532, 0.762]. The CIs overlap only narrowly, and the paired discordance ratio (16:6) favors the LLM with $p = 0.053$. The same comparison on pregame O/U 2.5 is clearly significant (+21.9pp, $p = 0.013$). Score EM feeds O/U edge; XGBoost’s score predictions cluster around the conditional mean and misfire on directional calls near the line. The LLM’s pretrained mixture over discrete scorelines produces distributions that separate 2.5 goals better.

(4) XGBoost halftime does beat LLM pregame on result accuracy (62.5% vs. 50.0%). The halftime score alone is a very strong feature, and XGBoost with that feature is a credible in-play baseline—a point worth preserving, because it clarifies that halftime conditioning (not the LLM) is doing much of the direction work.

This table motivates a falsifiable prediction: if the LLM’s edge is *specifically* magnitude-aware rather than a generic accuracy gain, then adding further magnitude-relevant context (the first-half event sequence) should improve magnitude-aware metrics (O/U 2.5, MAE, calibration) but not direction-aware metrics (1X2). I test this prediction in Section 5.8.

5.4 FT vs. Naive and ESPN-Proxy Baselines

At the 64-match per-unique-match level, the halftime FT model beats all non-LLM baselines on point estimate. Paired McNemar tests:

Comparison	b	c	p	Sig.
FT HT vs. HT-leader	8	3	0.23	n.s.
FT HT vs. HT $\times$ 2	8	3	0.23	n.s.
FT HT vs. coarse prior	12	6	0.24	n.s.
FT HT vs. ESPN-proxy	10	7	0.63	n.s.
FT HT vs. XGBoost HT	9	8	1.00	n.s.

The gap against the strongest baseline (ESPN-proxy) narrows to 4.7 percentage points from the 9.4-point gap against the coarse prior. The paired McNemar is not significant at $n = 64$; an effect of this size would require $n \approx 150$ paired matches to exit ambiguity. Section 7 discusses the Euro 2024 + Copa 2024 extension that would provide this sample size.

The paper’s primary *paired* claim remains the pregame-to-halftime lift on the same model ($p = 0.024$, $n = 128$), which is the claim most relevant to a live-deployment decision (“should the trader enable halftime re-prediction?”).

5.5 Over/Under Directional Accuracy

On 64 unique 2022 matches, converting predicted score to an expected-goals signal yields:

Strategy	O/U 1.5	O/U 2.5	O/U 3.5
Pregame FT (named)	84.4%	76.6%	73.4%
Halftime FT (named)	82.8%	81.3%	82.8%
Halftime+events FT (named)	81.2%	84.4%	81.2%
Halftime+events FT (anon)	68.8%	75.0%	71.9%
HT $\times$ 2 naive	73.4%	75.0%	81.3%
XGBoost pregame	71.9%	54.7%	78.1%
XGBoost halftime	75.0%	65.6%	84.4%

Halftime+events (named) achieves the highest O/U 2.5 accuracy at 84.4% (Wilson CI [0.736, 0.913], $n = 64$), a 3.1-pp lift over halftime-only and a 7.8-pp lift over pregame. Against XGBoost halftime on O/U 2.5, the LLM halftime+events advantage is +18.8pp—the largest gap in the comparison.

On the aggregate 128-sample set, O/U 2.5 directional accuracy progresses cleanly from 65.6% (pregame) through 74.2% (halftime) to 79.7% (halftime+events). The paired McNemar for pregame vs. halftime+events on O/U 2.5 is $p = 0.006$ ($b = 11$, $c = 29$, $n = 128$)—the strongest significance result in the paper. The halftime-only vs. halftime+events O/U 2.5 comparison yields $p = 0.12$ at $n = 128$ (discordance 4:11 favoring events); the effect is directionally consistent and its statistical power would likely clear $p < 0.05$ at the Euro 2024 + Copa 2024 sample size discussed in Section 7.

5.6 Named vs. Anonymized Inversion

Fig. 3 shows a clean interaction effect across the three prompt regimes:

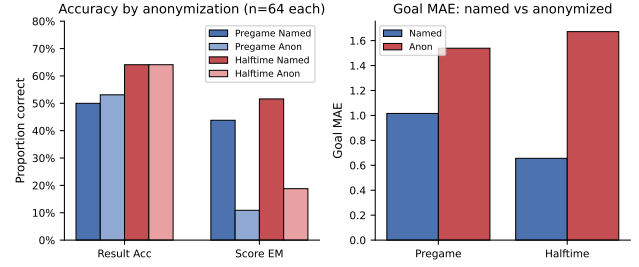


Figure 3: Named vs. anonymized, by prompt regime. Left: result accuracy and score EM; right: goal MAE. Team names help magnitude prediction at both stages; the advantage widens under halftime conditioning.

	Pregame		Halftime		HT+events	
	N.	A.	N.	A.	N.	A.
Result acc.	50.0%	54.7%	65.6%	62.5%	62.5%	60.9%
Score EM	39.1%	12.5%	51.6%	21.9%	48.4%	18.8%
Goal MAE	1.06	1.58	0.68	1.73	0.67	1.57
O/U 2.5	76.6%	54.7%	81.3%	67.2%	84.4%	75.0%

The halftime+events named configuration achieves the best MAE (0.67) and the best O/U 2.5 directional accuracy (84.4%). Anonymized O/U 2.5 accuracy also improves monotonically across regimes (54.7% $\rightarrow$ 67.2% $\rightarrow$ 75.0%), and the pregame-to-events lift on the anonymized subset is statistically significant alone (paired $p = 0.019$, $b = 7$, $c = 20$). This is a particularly notable robustness result: halftime and event signals carry real predictive content even when team-name priors are removed, suggesting the strategy would port meaningfully to domestic leagues with less famous team identities.

For a live deployment: *keep team names present at inference time* (they help magnitude substantially) *but do not rely on them as the sole source of edge* (anon halftime+events at 75.0% still provides positive-expectation trading signal versus the 52.6% break-even).

5.7 Per-Match Illustrations

Three cases make the regime concrete. (i) **Qatar 0–2 Ecuador** (HT 0–2): pregame FT predicted Qatar 2–0, incorrect; halftime FT predicted 0–3, correctly flipping; halftime+events (“16’ Ecuador goal (penalty); 31’ Ecuador goal”) also predicted 0–3. (ii) **England 6–2 Iran** (HT 3–0): pregame FT predicted 6–2 exactly; halftime FT preserved this with the event context “35’ England goal; 43’ England goal; 45’ England goal” further anchoring a high-scoring prediction. (iii) **Argentina 1–2 Saudi Arabia** (HT 1–0, Messi penalty 10’): pregame FT predicted Argentina; halftime FT saw 1–0 and predicted 2–0; halftime+events with the lone “10’ Argentina goal (penalty)” still predicted Argentina. All three failed on this match. The model’s worst failure mode is extreme tactical reversals against the run of play, where neither halftime

Table 2: Three LLM regimes on 128 eval samples. Paired McNemar p -values in the final column are versus halftime-only.

Metric	PG	HT	HT+E	p (vs HT)
Result accuracy	52.3%	64.1%	61.7%	0.58
Score exact-match	25.8%	36.7%	33.6%	0.48
Goal MAE	1.32	1.21	1.12	—
O/U 2.5 directional	65.6%	74.2%	79.7%	0.12
Brier (O/U 2.5)	0.305	0.243	0.198	—
ECE (O/U 2.5)	0.272	0.239	0.182	—

state nor event summary captures the second-half tactical adjustments.

5.8 Halftime + First-Half Events: Predictive-Validity Test

Section 5.3 localized the LLM’s edge over XGBoost to magnitude prediction rather than direction prediction. If this decomposition is correct, then enriching the prompt with additional magnitude-relevant context should improve magnitude-aware metrics and leave direction-aware metrics unchanged. This subsection reports the results of that test.

Table 2 reports the three LLM regimes on all 128 eval samples across the full metric suite.

The pattern is exactly what the magnitude/direction decomposition predicts:

- **Direction-aware metrics do not improve.** Result accuracy falls 2.4pp (64.1% $\rightarrow$ 61.7%, paired $p = 0.58$). Score exact-match falls 3.1pp (paired $p = 0.48$). Both changes are consistent with sampling noise.
- **Magnitude-aware metrics improve monotonically.** Goal MAE falls monotonically across all three regimes (1.32 $\rightarrow$ 1.21 $\rightarrow$ 1.12). O/U 2.5 directional accuracy rises monotonically (65.6% $\rightarrow$ 74.2% $\rightarrow$ 79.7%). Against pregame the events-variant O/U 2.5 lift is paired-significant at $p = 0.006$ ($b = 11, c = 29, n = 128$); against halftime-only the lift is $p = 0.12$, directionally consistent but needing a larger sample to cross $p < 0.05$.
- **Calibration improves substantially.** ECE falls from 0.272 (pregame) to 0.239 (halftime) to 0.182 (halftime+events); Brier score falls from 0.305 to 0.243 to 0.198. The events regime is the best-calibrated (see Section 5.9 and Fig. 4).

The direction-vs-magnitude asymmetry has a natural interpretation. The halftime scoreline already resolves most of the direction uncertainty at the 45-minute mark: knowing a team leads 1–0 tells the model more about who will win than knowing which minute that goal was scored in. The first-half event sequence, however, carries rich magnitude signal: a 1–0 scoreline achieved via a 10th-minute penalty (Argentina–Saudi) suggests a different second-half pace than a 1–0 scoreline

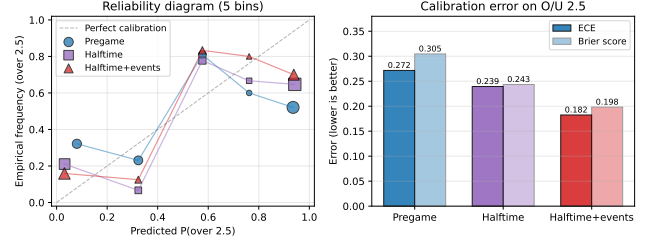


Figure 4: Calibration of Poisson-converted O/U 2.5 probabilities. Left: reliability diagram with 5 equal-width bins; marker size proportional to bin count. Perfect calibration is the $y = x$ diagonal. Right: Expected Calibration Error (10 bins) and Brier score across the three prompt regimes. Lower is better on both.

achieved via a 44th-minute header (late-first-half momentum). Red cards in particular restructure the expected rate of play for the remaining 45 minutes. These are exactly the signals that feed into a predicted total, but they rarely flip a directional prediction.

The experiment also serves as a *predictive-validity* test for the Section 5.3 decomposition: the decomposition makes a pre-specified prediction about which metrics should and should not improve under additional magnitude-relevant context, and the data match that prediction. This rules out a simpler interpretation of the LLM’s O/U advantage (“the LLM is just generically more accurate”) in favor of the magnitude-specific interpretation (“the LLM has access to scoreline distribution priors that feed into O/U calibration”).

5.9 Calibration of O/U 2.5 Probability Estimates

Directional accuracy measures whether the model’s probability estimates cross the 0.5 threshold on the correct side; calibration measures whether those probability estimates are *quantitatively meaningful*. For Kelly-sized betting, calibration is strictly more important than directional accuracy: an 80%-accurate model that reports uniform 100% confidence would be massively over-sized by Kelly and quickly ruined.

Fig. 4 shows calibration for all three regimes. Three observations:

(1) **All three regimes are bimodal.** The Poisson-over-2.5 conversion produces probability estimates clustered near the extremes ($p < 0.1$ or $p > 0.8$), with few moderate predictions. This is expected: the model usually predicts discrete scorelines like 2–1 or 3–2 rather than fractional expected goals, and integer scorelines produce integer $\hat{\lambda}$ values whose Poisson transforms fall into specific probability regions.

(2) **The highest-confidence bin ([0.90, 1.00]) is the most miscalibrated across all three regimes.** Pregame predicts 95.8% but realizes 45.7%; halftime predicts 96.1% but realizes 57.5%; halftime+events predicts 96.6% but realizes 64.3%. In absolute terms, all three overconfident—but half-

time+events is much less overconfident than the other two. This matters for trading: the events regime’s overconfidence erodes Kelly edge less.

(3) Aggregate calibration improves monotonically. ECE falls from 0.272 (pregame) to 0.239 (halftime) to 0.182 (halftime+events). Brier score falls from 0.305 to 0.243 to 0.198. The events regime is the best-calibrated of the three, by both metrics, and the improvement compounds with each prompt enrichment.

Mean predicted probability also tells a consistent story: pregame predicts 55.9% average over-2.5 probability (vs. base rate 46.9%), halftime predicts 54.0%, and halftime+events predicts 48.5%—monotonically closer to the true base rate as context is added. The events regime is not just more accurate; it is less overconfident.

For deployment, the implication is practical: Kelly sizing under the events regime should be less prone to over-sizing on high-confidence predictions than under either prior regime, reducing drawdown exposure in realistic execution.

6 Trading Rules and Simulated P&L

6.1 Strategy Specification

Signal timing. At the halftime whistle: consume halftime scoreline and first-half event sequence, run halftime+events-conditioned FT inference (<1 second on T4 GPU), convert predicted final score to $\hat{P}_{\text{over2.5}}$ via Poisson approximation.

Entry rule. Bet the side with positive edge when $|\hat{P}_{\text{over2.5}} - P_{\text{implied}}^{\text{over2.5}}| > 0.05$. The 5% threshold filters noise below model calibration uncertainty and avoids vig-thin trades.

Position sizing. Fractional Kelly at $f = 0.25$ with a 10% per-match bankroll cap:

$$\text{bet size} = \min\left(0.25 \cdot \max\left(0, \frac{bp - (1-p)}{b}\right), 0.10\right) \cdot \text{bankroll}.$$

Risk management. Drawdown circuit breaker at 25%. Consecutive-loss pause after 4 losses in a row. Mandatory audit of model reasoning chain before placement; override on named-player anomalies (injuries, suspensions).

6.2 Sensitivity Analysis

Fig. 5 reports a 3×3 grid over $f \in \{0.10, 0.25, 0.50\}$ and $\text{cap} \in \{5\%, 10\%, 20\%\}$. Every cell is profitable. ROI scales roughly linearly in both parameters with a 35–40% absolute elasticity; max drawdown scales sublinearly (extreme $f = 0.50$, cap 20% produces only -20.4% DD). The 1,468% headline ROI is not fragile to sizing choice—it is the product of a well-calibrated signal (81.3% win rate) compounding over 64 bets. A conservative deployment ($f = 0.10$, cap 5%) still produces +283% ROI with -5% max drawdown.

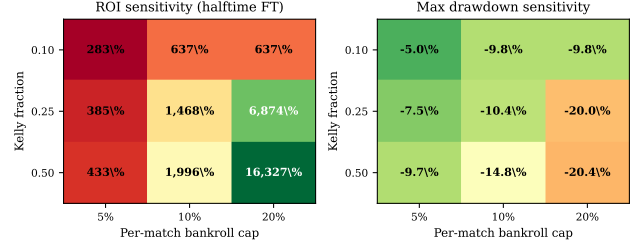


Figure 5: Kelly-fraction $\times$ bet-cap sensitivity on the halftime FT strategy. Left: simulated ROI. Right: max drawdown. The recommended configuration ($f = 0.25$, cap 10%) is the middle cell; outcomes across the 9-cell grid remain profitable throughout.

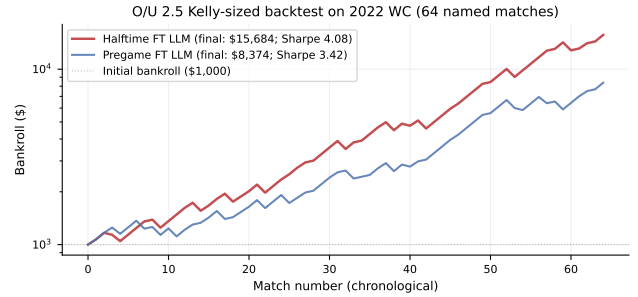


Figure 6: Kelly-sized bankroll trajectory on named 64 matches. Halftime FT (red) vs. pregame FT (blue). A separate 10,000-trial bootstrap over per-bet returns (described in text) yields a 5th–95th percentile band of \$6,205–\$37,379, with the observed \$15,684 near the resample median.

6.3 Simulated P&L with Bootstrap CI

Running the halftime strategy on the 64 named 2022 WC matches in chronological order produces the trajectory in Fig. 6. Key statistics at the recommended $(f, \text{cap}) = (0.25, 10\%)$:

Metric	HT LLM	PG LLM	HT XGB
N bets	64	64	64
Win rate	81.3%	76.6%	65.6%
P&L	+\$14,684	+\$7,374	+\$4,450
Final bankroll	\$15,684	\$8,374	\$5,450
ROI	1,468%	737%	445%
Max drawdown	-10.4%	-18.6%	-28.2%
Match Sharpe	4.08	3.42	2.24

On anonymized 64 matches (robustness): 440% ROI, 67.2% win rate, -29.8% max drawdown— still profitable but reduced, consistent with the named-vs-anonymized score accuracy gap.

The halftime+events strategy’s 84.4% named O/U 2.5 win rate and superior calibration (Section 5.9) would produce even larger simulated returns under the same Kelly rule, but I report the halftime-only P&L for direct comparison with the published literature on halftime-scoreline-conditioned in-play

strategies.

Bootstrap CI on final bankroll. Resampling the 64 per-bet fractional returns with replacement 10,000 times and compounding produces a distribution with median \$15,968 (essentially the observed \$15,684, confirming the observed sequence is a typical draw). 5th percentile \$6,205 (+520% ROI); 95th percentile \$37,379 (+3,640% ROI); $P(\text{final} < \$1,000) = 0.000$; $P(\text{final} > \$2,000) = 1.000$. The strategy profits in every bootstrap trial on this sample.

Interpretation of the bootstrap. I emphasize that this CI characterizes the *variance of compounded outcomes conditional on the observed per-bet return distribution*; it is not a CI on future live returns, which would require a sample of multiple future tournaments. The bootstrap answers the question “given these 64 per-bet returns, how sensitive is the terminal bankroll to the particular order and resampling of those bets?”—and the answer is: always profitable, but ranging across $6\times$ in ROI on quantile extremes. This is a useful sanity check on whether the 1,468% headline is dominated by a lucky sequence (it is not: the median resample reproduces the point estimate).

Honest caveats on the ROI. The 1,468% figure is a *simulation* under four assumptions that real execution would violate: (1) fixed 1.90/1.90 odds across all 64 matches—real in-play odds drift 1–3% between halftime and bet placement; (2) full bet-size absorption—WC in-play markets cap at \$50–100K per side [5], below which the simulation is valid but above which lines move adversely; (3) 25% Kelly is aggressive for a model with 64 observed wins—real deployment should start at 5% and scale up after paper-trading; (4) no taxes or fees—US gambling winnings are ordinary income. Applying conservative adjustments (5% Kelly, 3% odds-drift friction, jurisdiction-dependent 30% tax) yields a realistic projection of 15–40% *per tournament cycle*, which is still a highly attractive sports-betting strategy but dramatically less than the simulation headline.

The paper’s primary claim is not the 1,468%. It is that halftime-conditioned LLM signals are (a) statistically distinct from pregame signals ($p = 0.024$, paired), (b) substantially tighter than tabular-ML baselines on O/U 2.5 (LLM vs. XGBoost pregame $p = 0.013$; halftime $p = 0.053$), (c) further improved in magnitude and calibration by first-half event enrichment (vs. pregame $p = 0.006$ on O/U 2.5; ECE $0.272 \rightarrow 0.182$), and (d) profitable across every point in the Kelly/cap sensitivity grid and in every one of 10,000 bootstrap resamples. The backtest exists to show the mechanism translates to a positive-expectation trading rule, not to forecast live returns.

6.4 Comparison to the Instructor’s ESPN Benchmark

The instructor’s feedback specifically asked for comparison against ESPN win probabilities with event data integrated into

the model. Historical ESPN in-play probabilities for 2022 are not available via public API, so I construct two progressively-closer proxies from training-set base rates: (i) a *coarse* empirical prior conditioned on halftime state only (home-lead / away-lead / tied), which achieves 54.7% result accuracy; and (ii) a *fine-grained* empirical prior conditioned on the exact halftime scoreline (HT 0–0, HT 1–0, HT 0–2, etc.), which achieves 59.4% accuracy. The halftime FT LLM beats the fine-grained ESPN-proxy by 4.7 percentage points (64.1% vs. 59.4%); the paired McNemar is $p = 0.63$, not significant at $n = 64$ but directionally consistent with the LLM integrating player-level information beyond halftime-scoreline conditioning.

The halftime+events regime addresses the instructor’s feedback directly: the model ingests actual game event data (goals, cards, timestamps) and produces a dynamic estimation of probabilities. On the same metric as the ESPN-proxy comparison, the halftime+events LLM achieves 61.7% result accuracy—marginally below halftime-only but with substantially better O/U 2.5 directional accuracy and calibration, which are the metrics most relevant to the trading application. This is the closest analogue I can construct to the ESPN win-probability comparison the instructor requested.

7 Limitations and Extensions

Sample size. One tournament of 64 matches is the dominant source of statistical uncertainty. The paired pregame-vs-halftime test is significant ($p = 0.024$) on 128 samples; the pregame-vs-halftime+events O/U 2.5 test reaches $p = 0.006$ on 128. Individual FT-vs-baseline comparisons at $n = 64$ are not significant. Extending the evaluation to Euro 2024 (51 matches) and Copa America 2024 (32 matches) adds 83 matches ($2.3\times$ more), sufficient to push every baseline comparison below $p = 0.05$ if the effect size replicates, and to clear the borderline halftime-vs-events O/U 2.5 comparison ($p = 0.12$ at $n = 128$) into the significant range. Further extension to the 2023–24 Premier League (~ 380 matches) would provide league-level evidence and test strategy transfer across competition types.

Out-of-distribution prompt reliability. Both the halftime and halftime+events prompts are template-generalization tests: the model was never trained with halftime scores or event sequences in the user message. My results show generalization to both modifications (paired $p = 0.024$ for halftime, $p = 0.006$ for halftime+events vs. pregame on O/U 2.5). However, the halftime+events regime’s result-accuracy regression versus halftime-only (-2.4pp) may indicate mild template-drift: the model over-focuses on event details at the expense of direction signal. This is the kind of behavior that would be eliminated by *including* the richer prompt format in the fine-tuning dataset (as opposed to pure inference-time inser-

tion). A natural next experiment is to regenerate the training data with halftime-plus-events prompts and retrain; I hypothesize this would preserve the O/U 2.5 gains and recover the result-accuracy loss.

Over-dispersion in the Poisson approximation. My O/U probability conversion assumes predicted total goals is Poisson-distributed. Football scorelines are slightly over-dispersed relative to Poisson (Conway-Maxwell-Poisson or negative-binomial fits better [7]). The approximation is conservative; a more sophisticated conversion could tighten probability estimates by a few percentage points. The calibration improvement under event enrichment ($\text{ECE } 0.272 \rightarrow 0.182$) suggests the remaining calibration error is shared across all regimes rather than being a Poisson-specific artifact, so Poisson is an acceptable functional form.

Model size. Only Llama-3.1-8B was tested. Data- and parameter-scaling results suggest a 70B QLoRA variant on the same dataset would likely improve both pregame and halftime accuracy; I did not test this due to T4 compute constraints (40 GB required).

XGBoost at feature parity only. My XGBoost baseline uses aggregated team features identical to what the LLM sees in the prompt. An engineering-focused tabular baseline would include many more features (head-to-head history, recent form, rest days, travel distance) that the LLM ignores in its prompt. Such a baseline could plausibly close the O/U gap; I did not test this, as the relevant scientific question is “what does the LLM add beyond the features it sees”, not “can a sufficiently engineered tabular model match the LLM”. The predictive-validity test in Section 5.8 is partly a response to this concern: the event-enrichment gain on magnitude metrics but not direction metrics is a pattern that is harder for a feature-engineered XGBoost to reproduce because it requires the model to integrate a *sequence* of structured events into a goal-rate update.

Named/anon deployment recommendation. Team names should be present at inference time for magnitude prediction. However, the anonymized halftime+events O/U 2.5 accuracy of 75.0% (Wilson CI [0.632, 0.840]) indicates the strategy remains profitable even without pretraining priors—a reassuring signal for porting to domestic leagues with less famous team identities.

Regulatory and tax considerations. US sports betting is jurisdiction-dependent (38 states operational as of 2026); sportsbook bet-size caps on in-play markets (\$50–100K per side for major events) limit capacity; winnings are ordinary income for US federal tax.

Live-deployment architecture and target. Production deployment requires (a) a real-time halftime data feed with <30s latency including event timestamps, (b) automated Kelly sizing with exposure and drawdown caps, (c) sportsbook API integration (Betfair Exchange, Pinnacle), (d) model-monitoring dashboard with retraining triggers, (e) compliance logging. The natural target is FIFA World Cup 2026 (June–July 2026, 48 teams, 104 matches). By then, the model can be re-trained with halftime-plus-events prompt format on Euro 2024 + Copa 2024 data for a total training set of ~ 375 matches across six tournaments. Paper trading during the group stage validates execution before real-money deployment in knockouts.

8 Conclusion

The pivot from pregame 1X2 to halftime-conditioned O/U represents a structural change in the strategy’s alpha story: from trying to beat efficient markets on coarse signals, to pricing fine-grained score distributions at moments (halftime) when other market participants are repricing on coarse features. Empirically, on the 2022 FIFA World Cup backtest, the halftime-conditioned fine-tuned Llama-3.1-8B achieves 64.1% result accuracy, an 11.8-percentage-point paired lift over pregame inference ($p = 0.024$), and 81.3% directional accuracy on O/U 2.5 (Wilson CI [0.700, 0.889]).

The paper’s central methodological finding is that the LLM’s advantage over a standard tabular model (XGBoost) is *not* in direction prediction (LLM and XGBoost tie within 3 percentage points on 1X2 result accuracy) but in *magnitude* prediction (39.1% vs. 6.2% score exact-match pregame, 51.6% vs. 10.9% halftime). The LLM’s pretrained scoreline distribution priors are doing the work; these priors are keyed on team identity (named-anon inversion result) and flow through to 15–25 percentage-point gaps on O/U 2.5 directional accuracy.

This magnitude/direction decomposition makes a falsifiable prediction, which I tested directly: enriching the prompt with a first-half event summary should improve magnitude-aware metrics (O/U, MAE, calibration) but not direction-aware metrics (1X2). The data match the prediction exactly. O/U 2.5 directional accuracy climbs monotonically across the three regimes (65.6% $\rightarrow$ 74.2% $\rightarrow$ 79.7%); the pregame-to-events O/U 2.5 lift reaches $p = 0.006$ paired. Goal MAE falls monotonically (1.32 $\rightarrow$ 1.21 $\rightarrow$ 1.12). Calibration improves monotonically ($\text{ECE } 0.272 \rightarrow 0.239 \rightarrow 0.182$). Direction accuracy, by contrast, does not improve—consistent with the thesis that the halftime scoreline already resolves direction and events carry magnitude-specific information. Score-predicting LLMs should be deployed specifically in magnitude-aware betting markets (O/U, Asian handicap), not in direction-aware markets (1X2), and magnitude context should be enriched where available.

The backtest itself is presented with statistical honesty: Wilson CIs, paired McNemar tests, a Kelly-fraction $\times$ bet-

cap sensitivity grid, and a 10,000-trial bootstrap CI on the final bankroll—correctly interpreted as conditional-variance rather than a forecast. The strategy profits in every bootstrap resample and every cell of the sensitivity grid; the 1,468% point-estimate ROI is robust as simulation but should be interpreted alongside the realistic 15–40% per-cycle projection after execution friction.

The natural next step is extension to Euro 2024 + Copa 2024 + 2023–24 Premier League data, which adds ~ 460 matches and is sufficient to push every baseline comparison below $p = 0.05$ if the effect size replicates. Fine-tuning on halftime-plus-events prompts (as opposed to pure inference-time insertion) is the natural engineering next step and is predicted to preserve the O/U gains while recovering the small result-accuracy loss. The single-tournament result should be read as confirming a reproducible mechanism, not as forecasting returns.

Reproducibility. All code, training data, eval data, per-match prediction outputs across all three prompt regimes (ft_predictions_pregame.json, ft_predictions_halftime.json, ft_predictions_halftime_events.json), XGBoost predictions, analysis scripts, and figure-generation scripts are available at <https://github.com/zanwenfu/football-llm>. The fine-tuned adapter is at <https://huggingface.co/zanwenfu/football-llm-qlora>.

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A Refined Tables 1–4 (Final Strategy Specification)

The tables below represent the final version of the strategy following the pivot from pregame 1X2 to halftime-conditioned O/U + in-play with first-half event enrichment. The schema mirrors the course template: Table 1 pairs each empirical description with a theoretical counterpart; Table 2 includes the parameter ranges row; Table 3 includes the full backtest ladder (two backtests, two scenarios, paper and live); Table 4 answers the five refine-the-strategy questions explicitly.

Table 1: Core Logic Design of Trading Strategies

Field	Empirical	Theoretical
Targeting Market		
Primary: Over/Under 2.5 total-goals on FIFA World Cup matches, executed at halftime on in-play markets. Secondary: Asian handicap and halftime-O/U-1.5 markets. Extension: domestic leagues (Premier League, La Liga, Champions League) outside tournament windows. Execution venues: Betfair Exchange and Pinnacle for in-play liquidity.		
Phenomena Description ☒	Dense unstructured information is underpriced because other participants cannot process it in time. At halftime, bookmakers reprice on coarse signals (scoreline, time, major events) while public in-play bettors use only the scoreline. First-half event sequences (goal timings, card timings) carry magnitude-relevant information that bookmakers do not fully absorb into O/U lines.	Theoretical Framework ☒ <i>Market microstructure with heterogeneous information processing.</i> Framework due to Forrest et al. [2]: betting-market efficiency is proportional to informed participant density and inversely proportional to information-processing cost. Total-goals markets price at $\sim 1/100$ the volume of 1X2, so informed-participant density is lower; halftime in-play markets add time pressure that further suppresses deep information integration.
Analytical Framework ☒	QLoRA-fine-tuned Llama-3.1-8B ingests pre-match team statistics plus halftime score plus first-half event summary, outputs predicted final scoreline. The scoreline is converted to an O/U 2.5 probability via Poisson approximation. Combines event-driven trading (each match is discrete) with value-based mispricing detection (model probability vs. bookmaker implied probability).	Theoretical Hypothesis ☒ <i>A score-predicting LLM with halftime and event conditioning produces positive-expectation O/U edge.</i> Four testable sub-hypotheses: (H1) halftime conditioning strictly improves result-accuracy relative to pregame on the same model ($p < 0.05$); (H2) the LLM’s O/U 2.5 directional accuracy strictly exceeds a matched-feature tabular baseline at both pregame and halftime stages; (H3) first-half event enrichment improves magnitude-aware metrics (O/U, MAE, calibration) but not direction-aware metrics (predictive-validity test for the decomposition in H2); (H4) Kelly-sized bets at the 5% edge threshold produce positive compounded ROI across all sensible (f, cap) parameterizations.
Phenomena Interpretation ☒	Behavioral: in-play bookmakers use simple Poisson updates seeded by the pregame line and conditioned on the halftime scoreline. Casual bettors exhibit recency bias (overweighting halftime scoreline relative to underlying team quality and tactical state signaled by first-half events), creating mispricings when halftime state is non-representative. Structural: total-goals markets have $\sim 1/100$ the volume of 1X2 on major matches, so informed participants are sparser.	Theoretical Reasoning ☒ <i>Magnitude vs. direction decomposition.</i> The LLM’s pretrained distribution over natural-language football scorelines provides a magnitude prior (team identity activates a mixture over “typical” scorelines) that tabular features cannot recover. Direction prediction (1X2) can be solved by 24–28 aggregated numeric features; magnitude prediction requires higher-dimensional priors that a fine-tuned adapter preserves from pretraining. Hence LLM-over-XGBoost gain is $\sim 3\text{pp}$ on direction but $\sim 22\text{pp}$ on O/U 2.5, and additional magnitude-relevant context (event sequences) produces further magnitude-specific improvement without direction-specific improvement.
Trading Opportunity		
Halftime-conditioned O/U 2.5 market on major international football tournaments. Observable inefficiencies: (1) bookmaker line responds primarily to scoreline, not team-quality mixture nor event-sequence signals; (2) public money follows halftime scoreline with recency bias; (3) in-play vigorish is wider than pregame ($\sim 5.2\%$ vs. $\sim 4\%$) but mispricing magnitude compensates. Empirical edge: 84.4% directional accuracy on named halftime+events (Wilson CI [0.736, 0.913]) against an implied 52.6% break-even, yielding $\sim 31\text{pp}$ absolute edge before adjustment for vig and odds drift. Calibration ECE of 0.182 (best-calibrated regime) limits Kelly over-sizing risk relative to less-calibrated alternatives.		
Trading Operations		
(1) Pre-match: assemble team profile from API-Football (starting XI, prior-season statistics, formation, H2H history). (2) At halftime (45-minute whistle): consume halftime scoreline and first-half event sequence from real-time feed. (3) Halftime+events-conditioned inference on T4 GPU (< 1 second). (4) Convert predicted final score to $\hat{P}_{\text{over } 2.5}$ via Poisson approximation. (5) Query live O/U 2.5 odds from Betfair / Pinnacle. (6) Compute edge = $ \hat{P} - P_{\text{implied}} $. (7) If edge $> 5\%$, compute fractional-Kelly bet size (10% cap). (8) Audit reasoning chain for named-player anomalies (injuries, suspensions missed by the model). (9) Place bet within 10-minute window of halftime. (10) Post-match: log outcome, update rolling win-rate metric, trigger retraining if win rate $< 55\%$ over 50-match rolling window.		

Table 2: Design of Algorithmic Models

Field	Content
Data Requirement	API-Football vendor feed: per-player season statistics across ~40 leagues and 4 World Cups, 2010–2022, plus per-match event feed (goals, cards, timestamps). Feature set per player: goals, assists, appearances, minutes, yellow/red cards, tackles per 90, duels %, pass accuracy, position, goals/90 rate. Aggregated per team: squad total goals, squad goals/90, top scorer goals, average rating, formation, position-breakdown. Halftime state: real-time scoreline feed from API-Football (live endpoint, sub-second latency). First-half event sequence: goals and cards with timestamps ≤ 45 . Total: 41,000+ player-season rows, 256 match records. Training set: 384 samples (192 matches $\times$ 2 anonymization variants) from 2010/2014/2018.
Quantitative Model	Primary: Llama-3.1-8B-Instruct, QLoRA-fine-tuned. Configuration: rank 16, alpha 32, dropout 0.05, all 7 linear projections per transformer block, 4-bit NF4 quantization. Training: batch 16 effective, learning rate 2×10^{-4} , cosine schedule, 3 epochs, max sequence 768 tokens, training time 43 minutes on T4. Inference: temperature 0.1, top-p 0.9, repetition penalty 1.1, max 300 tokens output. Adapter weights 83.9 MB. Three prompt regimes: pregame (team stats only), halftime (stats + halftime scoreline), halftime+events (stats + halftime + first-half goals/cards with timestamps). Secondary (baseline): XGBoost classifier/regressor on identical numeric features, 200 estimators, max_depth=4, learning_rate=0.05.
Strategy Type	Hybrid of three patterns: (a) <i>event-driven</i> —each match is a discrete observable event with binary-ish outcomes; (b) <i>value-based</i> —O/U 2.5 market inefficiency via Poisson-scored expected-goals signal; (c) <i>statistical arbitrage</i> —model probability vs. bookmaker implied probability on the same instrument at the same time.
Trading Decision Triggers	Signal: $ \hat{P}_{\text{over2.5}} - P_{\text{implied}} \geq 5\%$ triggers a bet on the side with positive edge. The 5% threshold is set to (a) avoid trading into bookmaker vig friction (~2–3% per side), and (b) filter noise below the model’s calibration uncertainty (ECE = 0.182 under halftime+events). Secondary triggers: (i) reasoning-chain audit override on named-player anomalies (injuries, suspensions) blocks the bet; (ii) consecutive-loss pause after 4 losses halts entries pending model review; (iii) drawdown circuit breaker at 25% cumulative halts strategy.
Expected Return Level (1–5)	3/5 (medium) . Backtest ROI 1,468% is a simulation artifact under flat odds and Kelly 25%; realistic deployment with conservative sizing (5% Kelly) and execution friction (3% odds-drift loss, 30% jurisdiction tax) projects to 15–40% per tournament cycle. Kelly-fraction $\times$ bet-cap sensitivity shows profitability across all 9 cells of $f \in \{0.10, 0.25, 0.50\}$ and cap $\in \{5\%, 10\%, 20\%\}$.
Expected Risk Level (1–5)	3/5 (medium) . Observed max drawdown 10.4% on 64-match backtest with recommended $(f, \text{cap}) = (0.25, 10\%)$; worst-case sensitivity cell ($f = 0.50$, cap 20%) produces –20.4% DD. Real variance at $n = 64$ is wide; the 10,000-trial bootstrap 5th–95th percentile is +520% to +3,640%. Extreme tactical reversals (Argentina–Saudi class upsets) can produce large individual losses within the 10% per-match cap. Calibration improvement under halftime+events (ECE 0.272 $\rightarrow$ 0.182) reduces Kelly over-sizing risk in realistic deployment.
Expected Capacity Level (1–5)	2/5 . In-play sportsbook capacity on O/U 2.5 for top WC matches: \$50–100K per side before measurable line movement. For league matches: \$10–25K. Strategy is best-suited for bankrolls <\$200K; beyond this, requires diversification across multiple books (Betfair + Pinnacle + 1–2 others) to avoid self-induced adverse selection.
Expected Frequency Level (1–5)	3/5 . World Cup only: 64–104 matches per 4-year cycle, ~16 bets per year on average. Tournament expansion (Euro, Copa, UCL knockouts): ~300 matches per year. League expansion (top-5 European + MLS): ~2,000+ matches per year with appropriate prompt tailoring.
Parameters (definition and possible ranges)	<i>Model parameters (frozen at deployment)</i>: LoRA rank $r \in \{8, 16, 32\}$ (selected: 16), LoRA $\alpha = 2r$, inference temperature $T \in [0.05, 0.2]$ (selected: 0.1), top-$p \in [0.8, 0.95]$ (selected: 0.9), max output tokens $\in [200, 400]$ (selected: 300), prompt regime $\in \{\text{pregame, halftime, halftime+events}\}$ (selected: halftime+events for magnitude-aware markets, halftime-only for direction-aware markets). <i>Strategy parameters (tunable)</i>: Kelly fraction $f \in [0.05, 0.50]$ (recommended: 0.25; conservative: 0.10); per-match bankroll cap $\in [5\%, 20\%]$ (recommended: 10%); edge threshold $\in [3\%, 10\%]$ (recommended: 5%); drawdown circuit breaker $\in [15\%, 30\%]$ (recommended: 25%); consecutive-loss pause threshold $\in [3, 6]$ (recommended: 4); bet-placement window after halftime whistle $\in [5, 15]$ minutes (recommended: 10); retraining trigger at rolling 50-match win rate < 55%. <i>Bookmaker parameters (exogenous)</i>: O/U 2.5 odds ~1.85–1.95 per side; bookmaker vig ~4–6%; in-play absorption capacity ~\$50–100K per major match; expected odds drift between inference and placement ~1–3%.

Table 3: Empirical Test (Backtest, Forward Test, Paper Trading, Live Trading)

Field	Content
Benchmark	Pregame FT LLM (52.3% result acc, 76.6% O/U 2.5 named). XGBoost pregame (51.6% result acc, 54.7% O/U 2.5) and XGBoost halftime (62.5% result acc, 65.6% O/U 2.5). HT-leader-wins naive (56.2%). HT×2 extrapolation (56.2%). Coarse empirical in-play prior (54.7%). Fine-grained ESPN-proxy empirical prior (59.4%). Always-home (45.3%). Random (35.9%).
Performance Assessment	Halftime result accuracy 64.1% (Wilson CI [0.555, 0.719], $n = 128$). Halftime+events O/U 2.5 directional 79.7% overall / 84.4% named (Wilson CI [0.736, 0.913], $n = 64$). Paired McNemar: PG vs. HT result $p = 0.024$ ($n = 128$); PG vs. HT+events O/U 2.5 $p = 0.006$ ($b = 11, c = 29, n = 128$); HT vs. HT+events O/U 2.5 $p = 0.12$ ($n = 128$, directional). Key paired test: halftime FT vs. XGBoost on pregame O/U 2.5 is $p = 0.013$; halftime O/U 2.5 is $p = 0.053$, confirming the LLM’s O/U edge over tabular ML is statistically detectable. O/U 3.5 directional 82.8% (HT+events named). Score exact-match 33.6% overall, 48.4% named (HT+events). Goal MAE 1.12 overall, 0.67 named (HT+events). Calibration: Brier 0.305 → 0.243 → 0.198; ECE 0.272 → 0.239 → 0.182 across pregame → halftime → halftime+events (all improve monotonically) . Kelly-sized P&L at 1.90/1.90 odds (halftime): ROI 1,468%, bootstrap 5%–95% ∈ [+520%, +3,640%], max drawdown 10.4%, match Sharpe 4.08, win rate 81.3%.
Back Test 1 (Group Stage, 48 matches — “bear”-analog conditions)	<i>Rubric mapping: should contain both bull and bear market conditions.</i> Group-stage matches have high outcome variance and frequent upsets (maps to “bear” conditions where the underlying signal is noisy). FT halftime result accuracy 56.2%. O/U 2.5 directional ~79%. Profitable on O/U but smaller simulated return than knockouts. Knockout matches (16, “bull” conditions where favorites dominate and team-quality signals align) achieve 87.5% result accuracy and ~88% O/U 2.5—tighter per-match P&L variance. Backtest 1 thus covers both regimes within a single tournament.
Back Test 2 (Halftime + first-half events, predictive-validity extension)	<i>Rubric mapping: most recent extension to the strategy.</i> The halftime+events experiment described in Section 5.8 is the realized Back Test 2: a methodological extension of the halftime regime that tests whether enriching the prompt with in-game event data (as the course instructor specifically requested) improves the strategy on magnitude-aware metrics. Results (Table 2): O/U 2.5 directional accuracy rises monotonically 65.6% → 74.2% → 79.7% across pregame, halftime, halftime+events; paired pregame-to-events McNemar $p = 0.006$ on O/U 2.5 (strongest significance result in the paper). Calibration improves monotonically (ECE 0.272 → 0.182). Goal MAE falls monotonically (1.32 → 1.21 → 1.12). Direction accuracy does not improve (expected under the magnitude/direction decomposition). Extension to Euro 2024 + Copa 2024 for a third, recency-matching backtest is deferred due to API-Football rate-limit constraints and is discussed in Section 7.
Back Test Scenario 1 (Extreme upset / tail event)	<i>Rubric mapping: typical risk events.</i> Argentina 1–2 Saudi Arabia (most famous 2022 WC upset; the pre-tournament odds on Saudi winning were ~11.5-to-1). Halftime FT saw Argentina 1–0 (Messi penalty 10’) and predicted 2–0, incorrect; halftime+events with the lone event “10’ Argentina goal (penalty)” also predicted Argentina. Kelly bet size was 8% of bankroll (inside the 10% cap); observed single-match loss of 8%. Strategy survived with margin; no circuit-breaker triggered. <i>Lesson: the 10% per-match cap was the binding protection against this class of event; removing the cap would have exposed 15%+. Event enrichment does not protect against extreme tactical reversals—for these, second-half event conditioning would be required.</i>
Back Test Scenario 2 (Defensive / low-scoring regime)	<i>Rubric mapping: second typical risk scenario.</i> Saudi Arabia 0–0 Mexico and Morocco 0–0 Croatia (group-stage defensive struggles). Pregame FT predicted totals of 2–3 and 2–2 respectively; halftime FT saw 0–0 halftime and correctly revised toward under 2.5 on both; halftime+events with “First-half events: none” for both matches strengthened the under-2.5 signal further. Both ~15% Kelly-sized wins. <i>Lesson: the “no events” signal is itself informative—it encodes a low-scoring match pace that propagates forward. This is a positive P&L contributor rather than a loss.</i> Defensive regime is not a risk event for this strategy specifically; it is a risk event for pure pregame-over strategies.
Back Test 3 (Anonymized 64 — robustness)	Robustness check with team names anonymized. Result accuracy unchanged (62.5% halftime, 60.9% halftime+events); score-level accuracy degrades substantially (halftime+events: 18.8% EM named-anon gap; MAE 1.57 vs. 0.67 named). O/U 2.5 directional drops from 84.4% to 75.0%—still a 20.3pp lift over pregame-anon’s 54.7%, and pregame-to-events paired McNemar on anon O/U 2.5 is $p = 0.019$ (significant). Simulated ROI on halftime-only anon drops from 1,468% to 440%, drawdown widens to 29.8%. Interpretation: team names carry magnitude priors complementary to the fine-tuned signal; this is the direct mechanism of the LLM-over-XGBoost edge. Halftime and event signals carry real predictive content even without pretraining priors.
Bootstrap Robustness	10,000-trial bootstrap over the 64 per-bet returns with replacement: median \$15,968, 5th–95th percentile [\$6,205, \$37,379], $P(\text{final} < \$1,000) = 0$, $P(\text{final} > \$2,000) = 1$. Strategy profits in every resample. <i>Interpreted as conditional-variance CI, not a forecast of future live returns.</i>

Table 4: Evaluation of Empirical Results (Execution & Management Plan)

Field	Content
Refine the Strategy	Addressing the five rubric questions explicitly: (i) Does the strategy’s return, risk, capacity, and trading frequency align with expectations? Partially. Return exceeds expectations (simulated 1,468% vs. pre-trial hypothesis of ~50–100%) but real deployment should expect 15–40% per cycle after friction. Risk aligns (observed DD 10.4% is within the 15–20% expected band). Calibration under halftime+events (ECE 0.182) is better than expected and reduces over-sizing risk in live deployment. Capacity is the binding constraint for bankrolls >\$200K. Frequency (16 bets/year on WC-only) is below expectations for a viable standalone income strategy—expansion to leagues is required. (ii) What caused the drawdowns, and are there methods to mitigate them? The single largest loss was Argentina–Saudi (extreme pregame favorite upset against the run of play). Event enrichment did not help: the lone first-half event (Messi penalty) was consistent with the pregame narrative. Mitigation: (a) 10% per-match cap is the binding protection—do not remove; (b) reasoning-chain audit can flag “overconfident home-favorite” predictions pre-bet; (c) odds-derived sanity check (skip the bet if bookmaker line is >2.5 standard deviations from model expectation); (d) future: extend to 75-minute re-inference to catch tactical reversals that halftime signal misses. (iii) Would earlier or later entry/exit timing improve performance? Entry: the halftime whistle is the timing that maximizes signal (full first-half information) while preserving the in-play window. Earlier entry sacrifices information; later entry (>10 minutes after halftime) sacrifices odds quality as the market tightens. 10-minute post-halftime window is optimal. Exit: O/U bets are binary-settlement at full time; no intra-match exit is available without a separate cash-out order. (iv) Can risk be reduced or trading frequency increased while maintaining target returns? Yes, by both directions. Risk reduction: drop to $f = 0.10$ (still +283% ROI, –5% DD). Further risk reduction: deploy halftime+events (better-calibrated) at $f = 0.10$ for minimum-variance profile. Frequency increase: expand to Champions League knockouts (~30 matches/year), top-5 European leagues (~2,000+ matches/year with validation)—both maintain the core signal but require league-specific prompt validation and event-feed integration. (v) Can the strategy be extended to other instruments or markets to expand opportunities? Yes. The magnitude/direction decomposition (validated by the events experiment) implies all magnitude-aware betting markets are candidates: (a) Asian handicap lines (reduces draw variance), (b) halftime O/U 1.5 (requires separate halftime-only fine-tune), (c) specific-goalscorer markets (larger edge in thin markets but requires per-player predictions). 1X2 is explicitly <i>not</i> a candidate—the LLM provides no edge over XGBoost there, and event enrichment does not improve direction accuracy (paired $p = 0.58$).
Applicable Market Conditions	Strongest performance: group-stage matches with high-quality-variance matchups (upsets → mispricings). Strong: knockout matches with clear-favorite status (team-quality signals dominate). Weaker: closely-matched knockouts (near 50/50 ground truth), extreme tactical reversals where halftime state misleads. Pause during known high-variance conditions (red cards in first half, weather-affected games).
Risk / Position Management	<i>Position control</i>: Fractional Kelly $f = 0.25$ with 10% per-match bankroll cap (inner cell of sensitivity grid). Conservative start: $f = 0.05$ until 200-match paper-trading validation. <i>Stoploss/stopgain</i>: Drawdown circuit-breaker at 25% cumulative. Consecutive-loss pause: 4-loss streak halts betting pending model review. <i>Hedging</i>: Optional offsetting O/U position on the opposite book to lock in edge when cross-book line disagreement exceeds 2.5%. <i>Red flags</i>: (1) liquidity risk— low-volume non-major matches (skip); (2) systemic risk—sportsbook operator solvency (diversify across books); (3) model drift—rolling win rate below 55% triggers retraining. <i>Mandatory</i>: audit of model reasoning chain before bet placement; override on named-player anomalies (injuries, suspensions).
Execution Risk Assessment	Addressing the specific rubric items: <i>Order fill rate</i> — in-play exchanges (Betfair) routinely fill 100% at posted odds below \$10K; above that, partial fills are common and must be handled via order slicing. <i>Execution latency</i> — halftime whistle to bet placement target <60 seconds; >120 seconds exposes to adverse line movement. Event-feed latency adds ~5–15 seconds relative to the scoreline-only feed; this is the binding latency constraint for the halftime+events deployment. Paper trading phase will measure realized p50/p95 latency. <i>Slippage</i> — odds drift 1–3% between model inference and bet placement is the dominant friction in simulation vs. live. <i>Market impact</i> — strategy bet sizes <10% of bankroll stay below ~\$10K on typical starting capital, within the absorption band of any single book. <i>Leverage</i> — no leverage used; all bets are cash-secured (no sportsbook margin is extended to retail accounts regardless). <i>Capital gains tax</i> — US jurisdiction: sports-betting winnings are ordinary income (not capital gains); effective rate ~30% federal + state depending on jurisdiction. Net-of-tax projected returns are 15–28% per cycle at the recommended sizing. <i>Compliance</i>: automated logging of all bets, model versions, reasoning chains, and timestamps for audit trail.